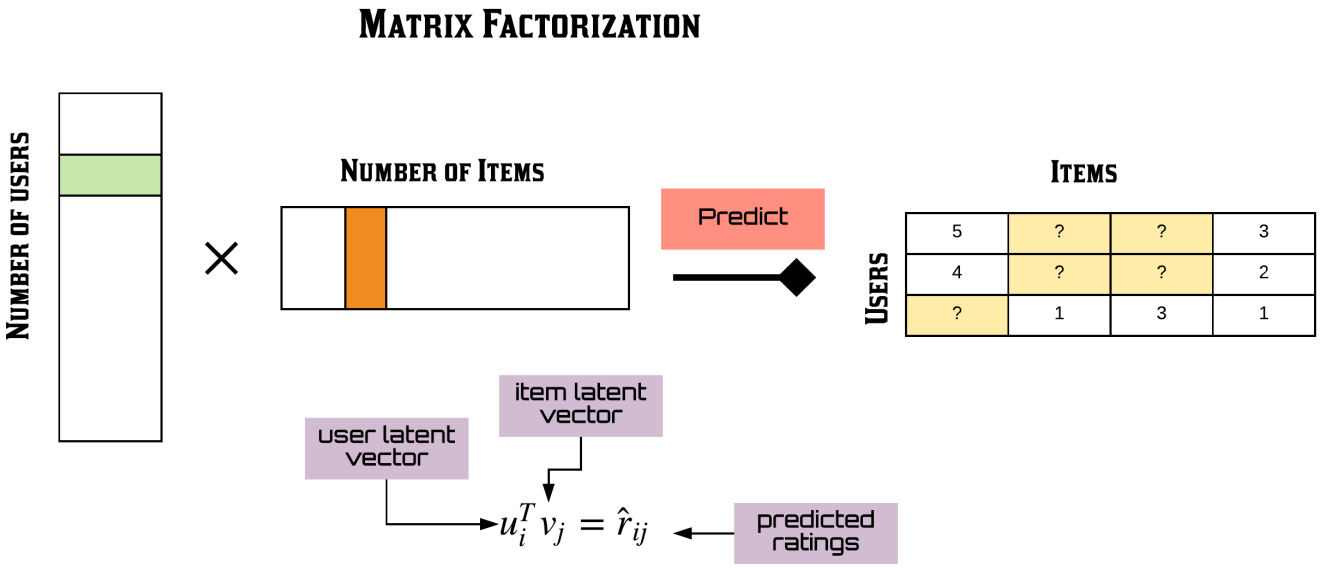
Recommend Locally, Diversify Globally

FedFlex for Live Federated Netflix Recommendations

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Emma Beauxis-Aussalet,
Manel Slokom

FLICS, Hybrid
09-06-2026

The Story



- Every day, billions of users consume recommendations from Netflix, Spotify, YouTube, TikTok
- None of them choose the algorithm
- None of them control what data is collected
- None of them understand why certain recommendations appear

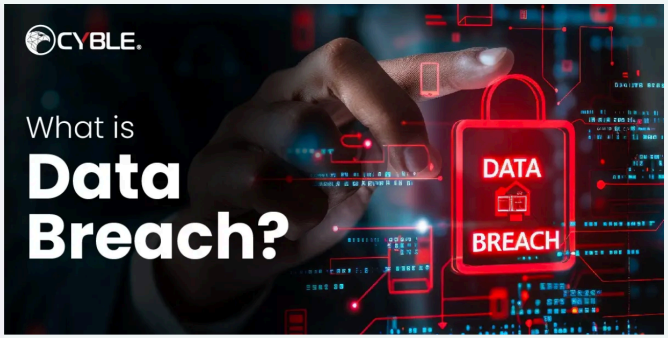
What if this could be different?

Credits: https://levelup.gitconnected.com/how-does-netflix-recommend-k-dramas-for-me-matrix-factorization-34f22d2a1c13

The Problem: Centralization as default

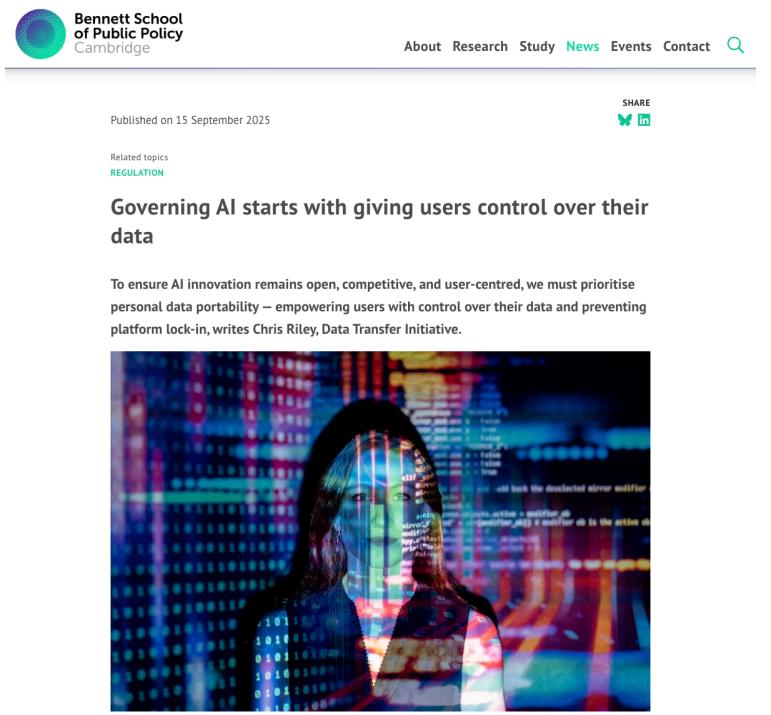
Privacy Crisis

- Consumption patterns mined continuously
- Users don't own their behavioral data
- No opt-out without losing service



Governance Crisis

- Algorithms optimize for company goals
- Diversity, fairness bolted on as afterthoughts
- “Governing AI starts with giving users control over their data”



Source:
https://www.bennettschool.cam.ac.uk/blog/governing-ai-starts-with-giving-users-control-over-their-data/
https://cyble.com/knowledge-hub/what-is-a-data-breach/

Why nobody has shown this works

- Federated learning literature: assumes perfect networks and rational agents
- User studies: assume people understand privacy-preserving systems
- Prior work: simulates federated learning or tests on offline datasets
- Never with real humans making real choices in real time

We asked the hard questions:

- Can we actually build this?
- Will real users engage with it?

Research questions

- **RQ1:** Can federated learning work with real users making real choices?
- **RQ2:** Do users prefer and trust systems where their data stays locally private?
- **RQ3:** When given algorithmic choice, what do users prefer: personalization or diversity?
- **RQ4:** How do we design interfaces that make privacy and governance visible?

System design: Three principles

Principle 1: Data Sovereignty

- Local training
- Secure aggregation
- Differential privacy (ε = 1.0, Gaussian noise, local DP)

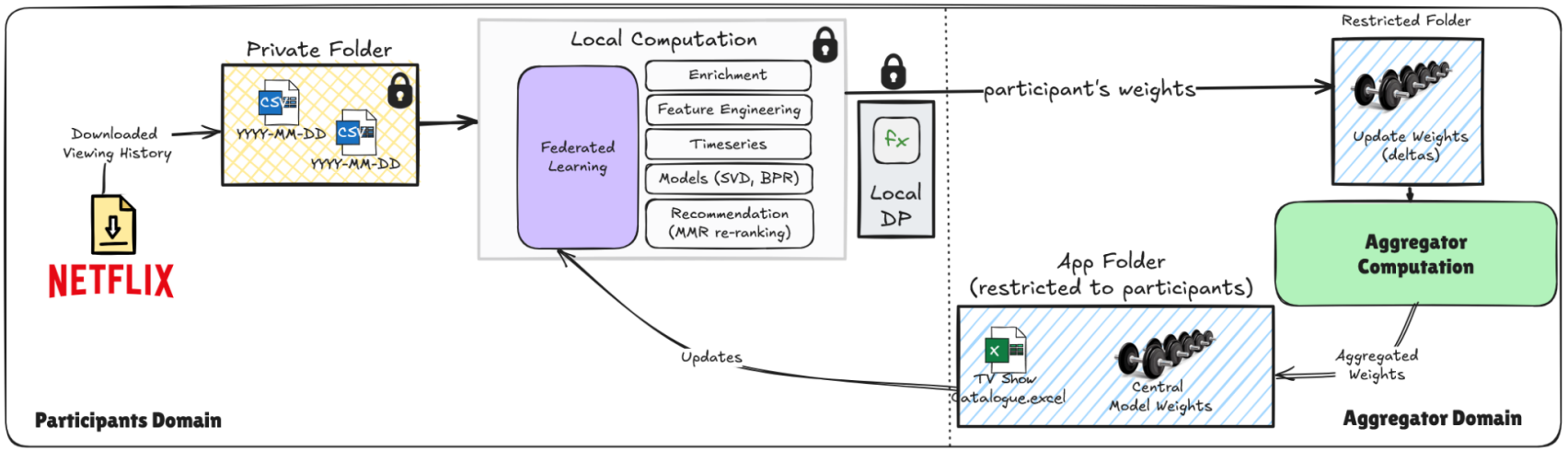
Principle 2: Transparency through interface

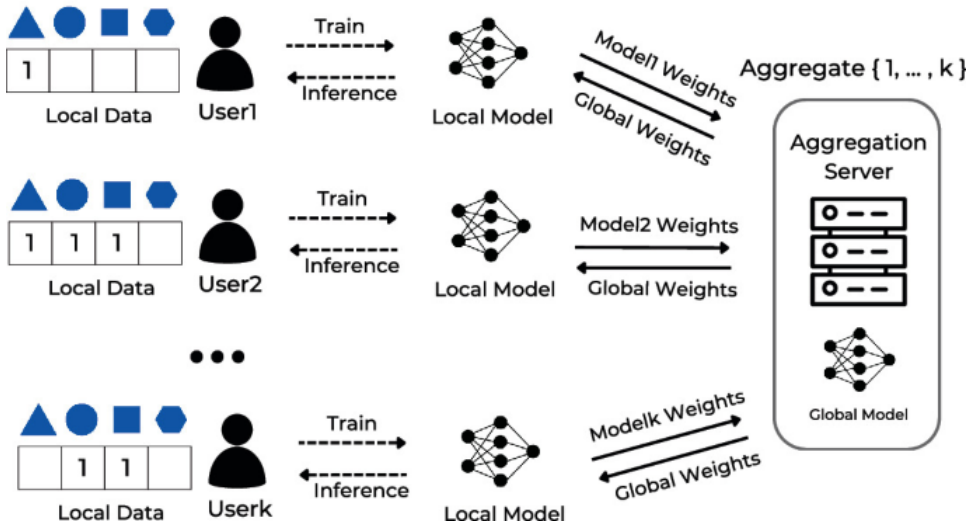
- MMR re-ranking (λ=0.3): post-hoc diversity, federated training unchanged
- Models: SVD (explicit feedback) and BPR (pairwise implicit)

Principle 3: Genuine agency

- Real choices with real consequences
- System responds immediately

Overal architecture

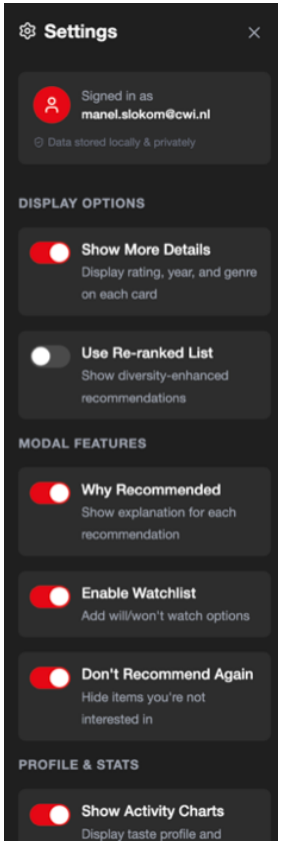




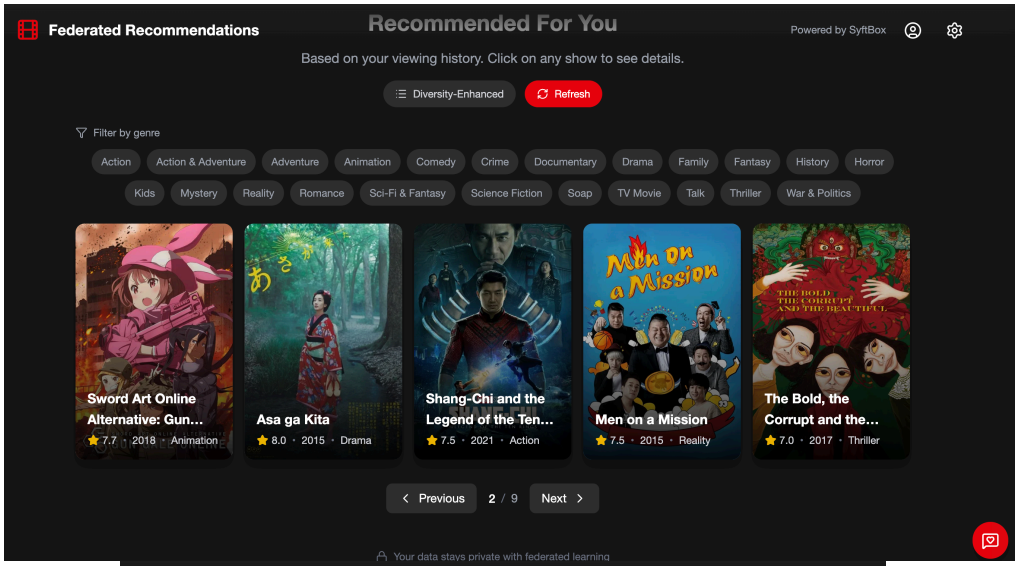
The Web Interface: The participants

- Netflix-like feed with local storage (no login required)
- Two modes: Standard or Diversity-Enhanced recommendations
- Immediate feedback on all user choices
- Settings panel with 7 feature toggles

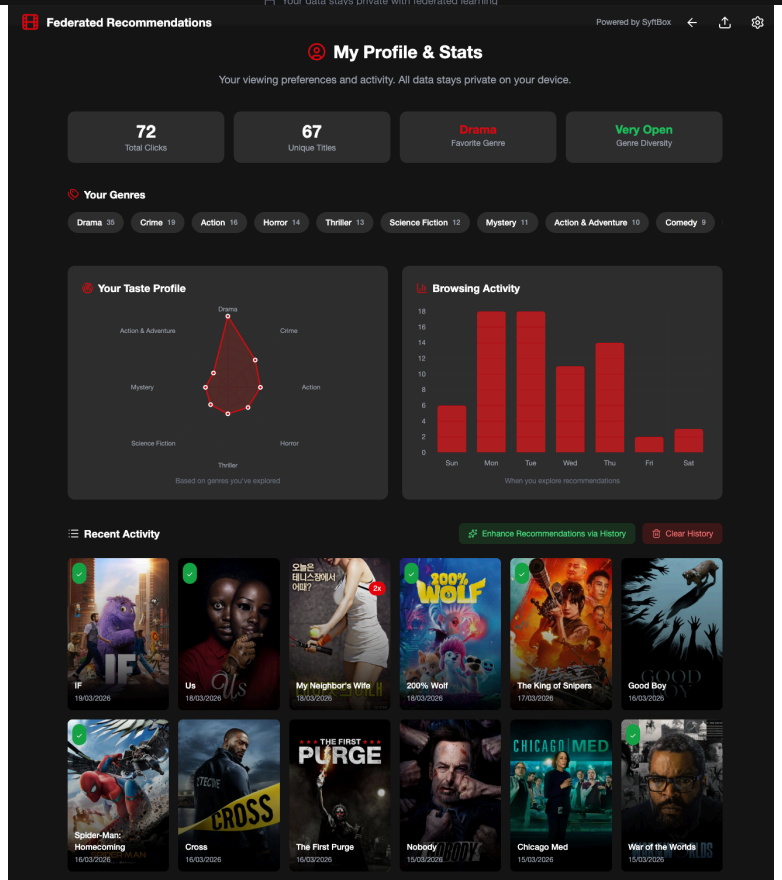
Why this matters: When architecture manifests control, users understand it without technical lectures.



(b) Settings

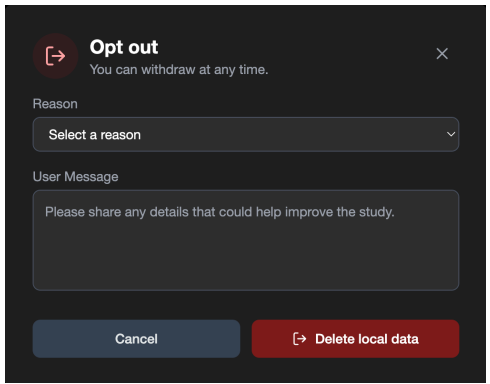


(a) Participant recommendations



(d) Participant's local profile & stats

(c) Opt-out box

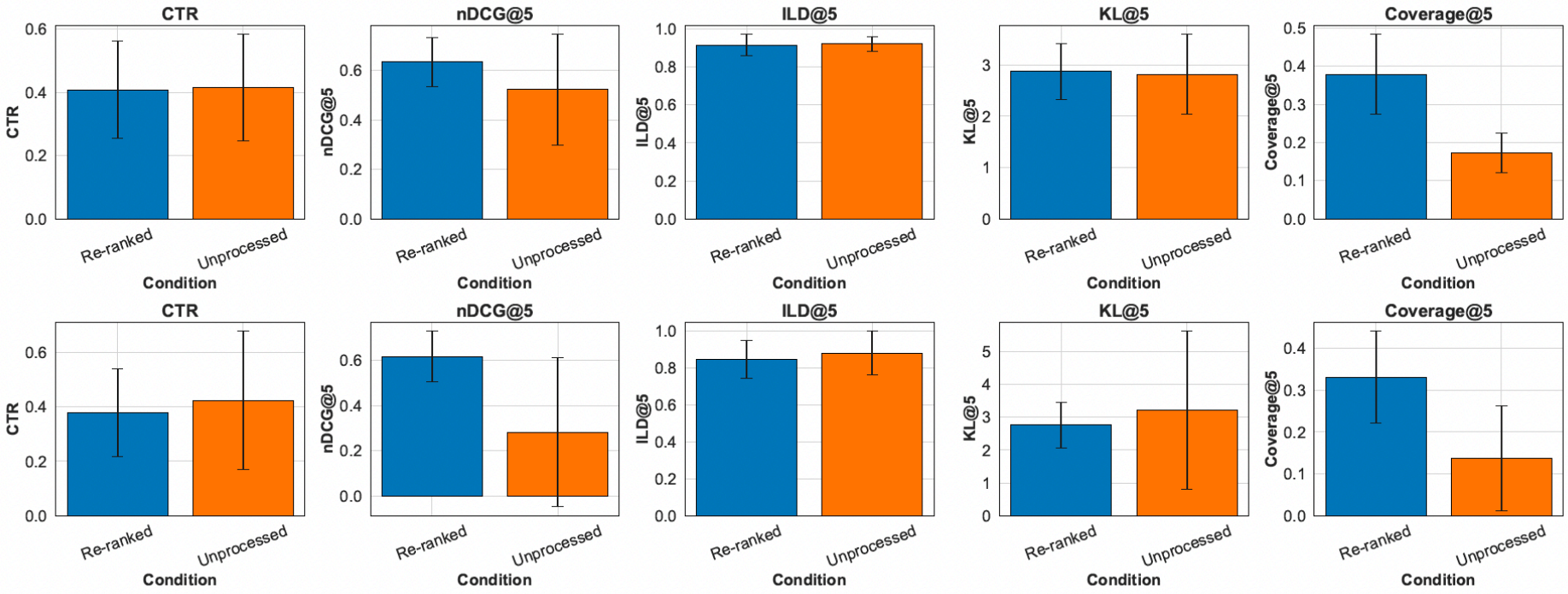


Participant Interfaces

Study design: The aggregator

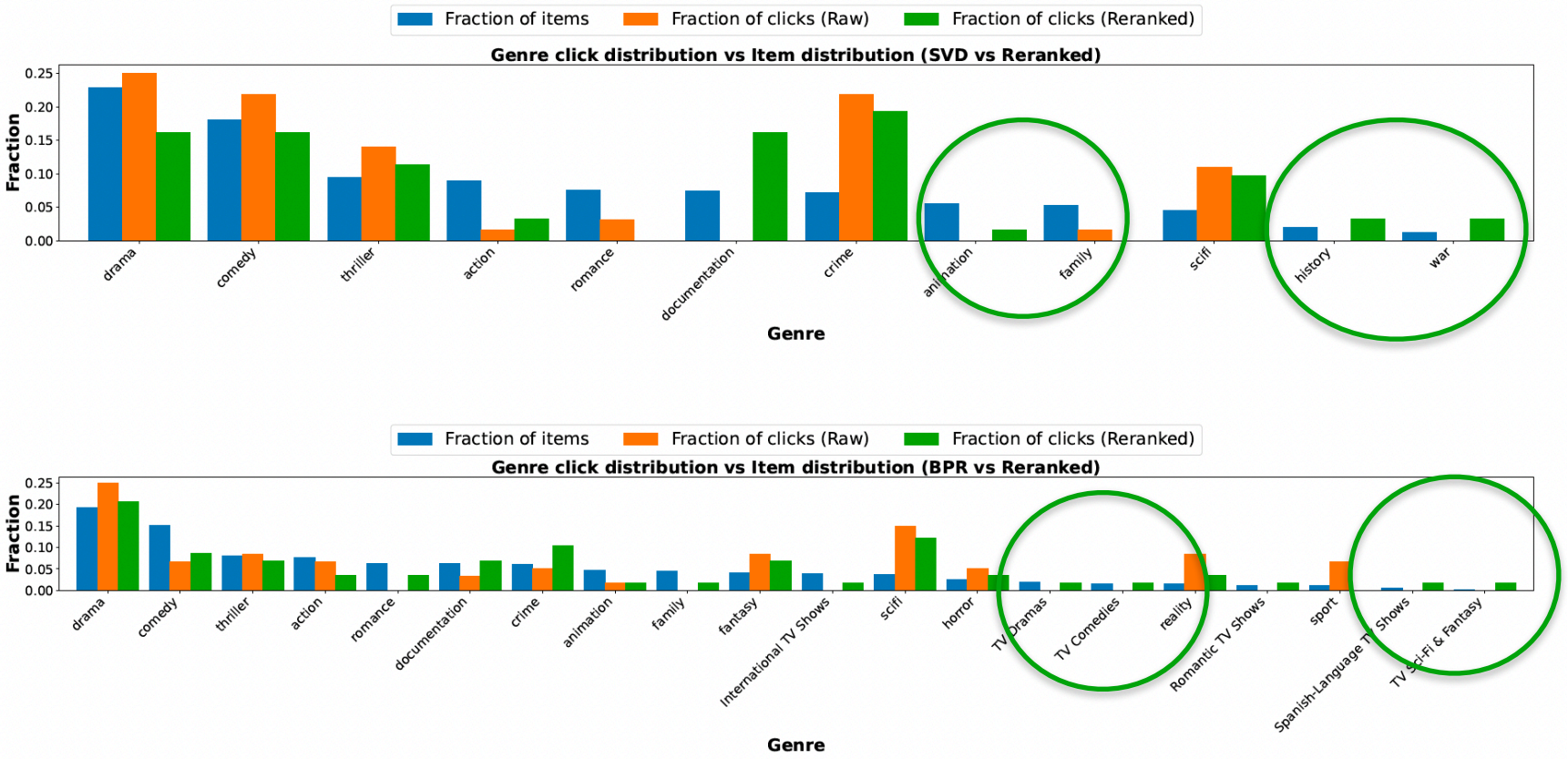
- **Participants**: 13 participants (2-week live deployment, via personal networks)
- **Duration**: 2 weeks; 10 participants completed full study + exit questionnaire
- **Data**: Netflix catalog with 8,807 titles across 40+ genres (Movies, TV series)
- **Conditions**: SVD vs BPR, each ± MMR re-ranking (top-5 recommendations)

BPR + MMR boosts ranking quality (nDCG +0.336*)



MMR re-ranking substantially increased Coverage Ratio for both SVD and BPR. Diversity gains did not reduce user satisfaction.

Distribution of clicked genres between raw and reranked



Re-ranked BPR consistently introduces clicks for new genres while still respecting strong user preferences (e.g., drama)

The Federated Learning Advantage

In Centralized Systems:

- Users wait for company to redesign algorithm
- No local recovery option

In Federated Systems:

- Users can refresh their local model
- Feedback immediately improves their recommendations

What users actually said

3/5: "After fine-tuning I get the same suggestions. It would be **strange** to continue after explicitly hiding it."

4/5: "In the **enhancement** I found Hotel Transylvania 3 and Jujutsu Kaisen. The **interface was useful**."

4/5: "In the **third** session there was already **improvement** in the genres suggested."

5/5: "**Great**, I got items I want to watch from the **first** recommended items..."

Interpretation: Even critical feedback was constructive, genuine and continuous engagement.

“User-controlled, privacy-preserving AI

isn't a future state.

It's a design choice we can make today.”

When you design for control rather than extraction, users respond. They engage. They understand. They prefer it.

Thank You

Let's Give Control Back to Users

Thank you to the our participants for trust, time and unlimited support!

Questions?